

Assignment-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

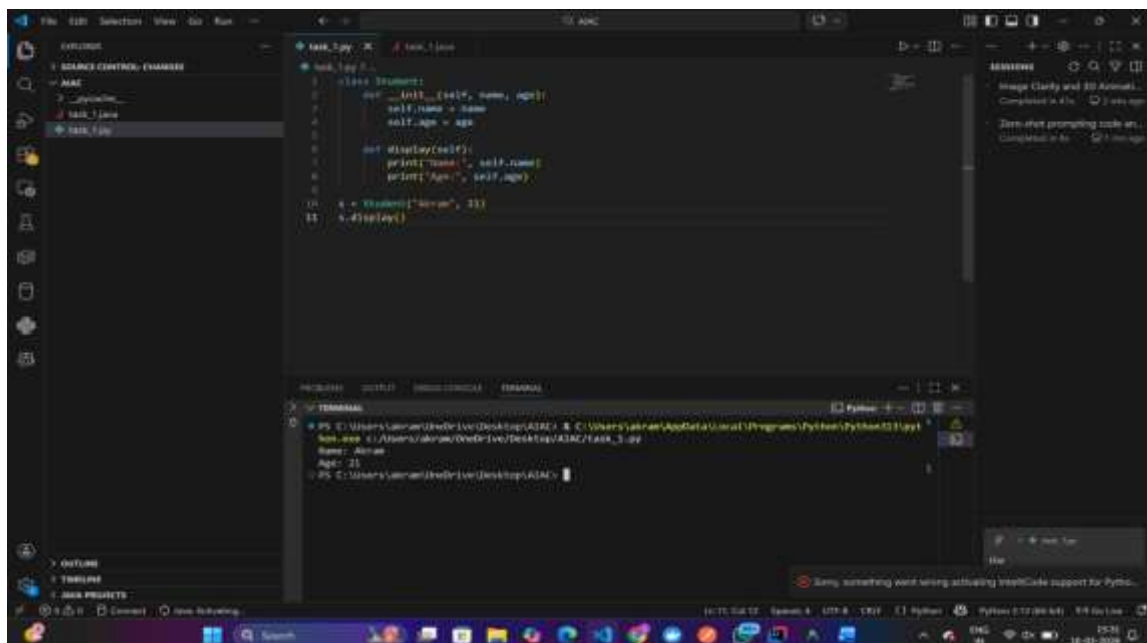
Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

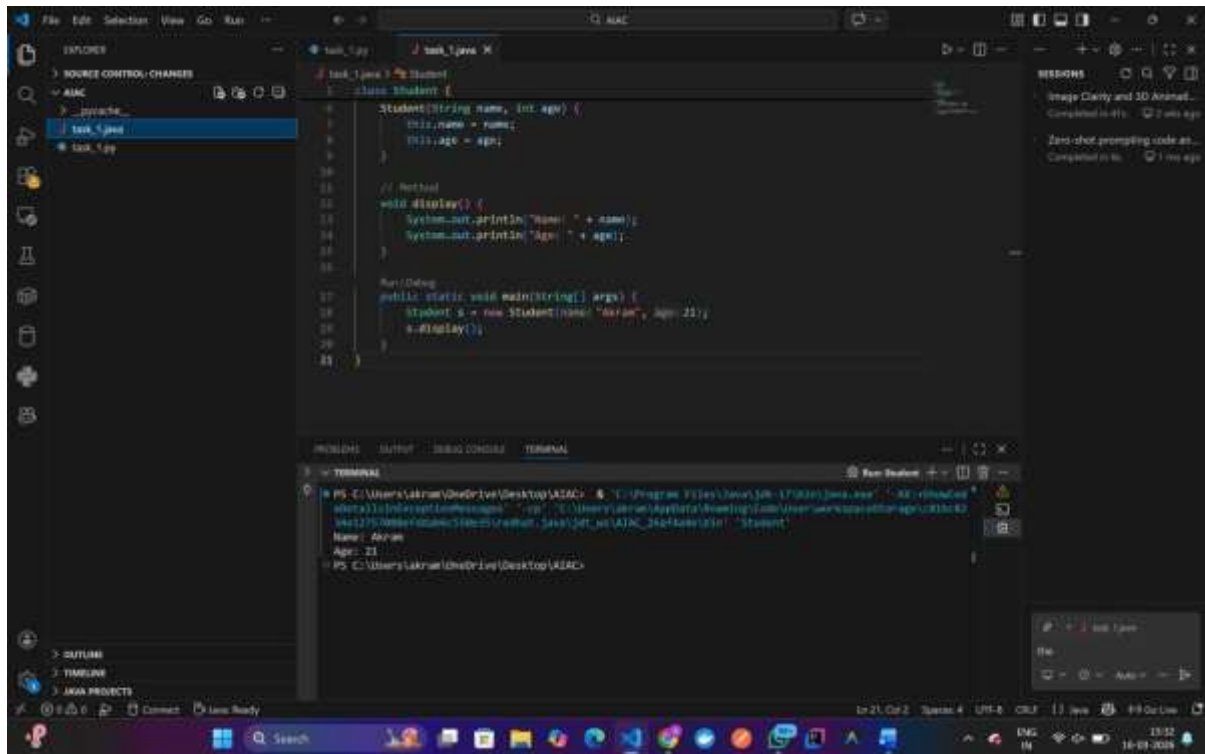
CODE:



The screenshot shows an IDE with a Python file named 'task_1.py'. The code defines a 'Student' class with attributes 'name' and 'age', and methods 'display_name' and 'display'. An instance 's' is created with 'Sujal' as the name and 21 as the age, and the 'display' method is called. The terminal output shows the execution of the script, resulting in 'Name: Sujal' and 'Age: 21'.

```
task_1.py
1 class Student:
2     def __init__(self, name, age):
3         self.name = name
4         self.age = age
5
6     def display_name():
7         print("Name:", self.name)
8         print("Age:", self.age)
9
10 s = Student("Sujal", 21)
11 s.display()
```

```
PS C:\Users\laksh\Desktop\ADAC> python task_1.py
Name: Sujal
Age: 21
PS C:\Users\laksh\Desktop\ADAC>
```



Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

- Ask AI to translate a recursive C++ factorial function into Python.
- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

CODE:

```

def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)

print(factorial(5))

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python113\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task1.py
Name: Akram
Age: 23
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python113\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

```

def factorial(n):
    result = 1
    for i in range(1, n+1):
        result *= i
    return result

print(factorial(5))

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python113\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task2.py
Name: Akram
Age: 23
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python113\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task3.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

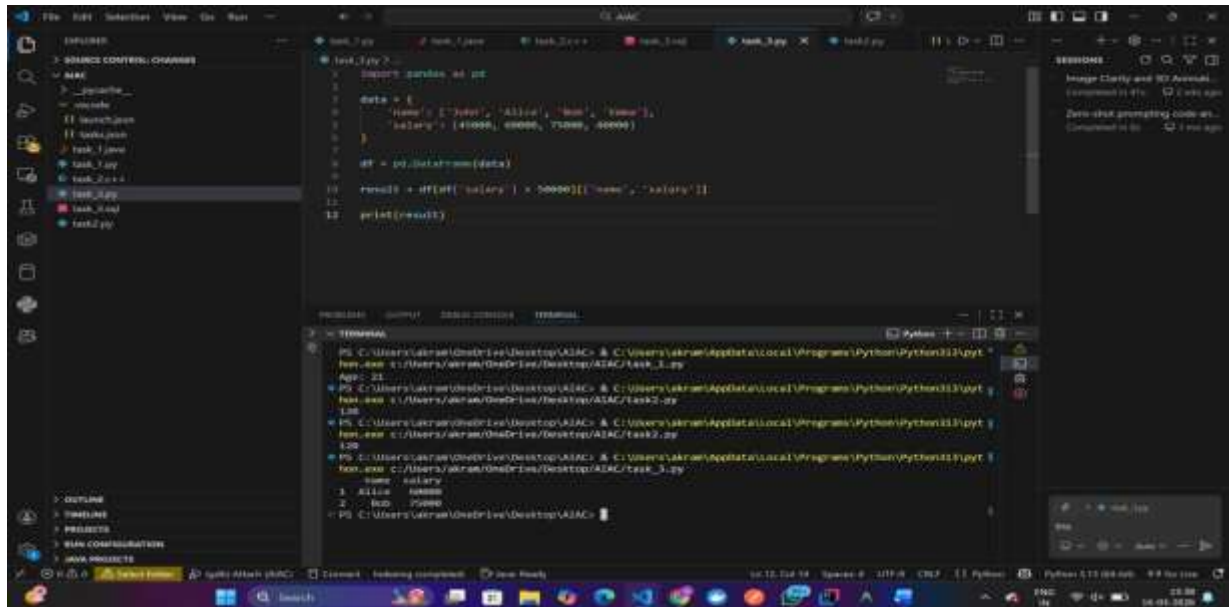
Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

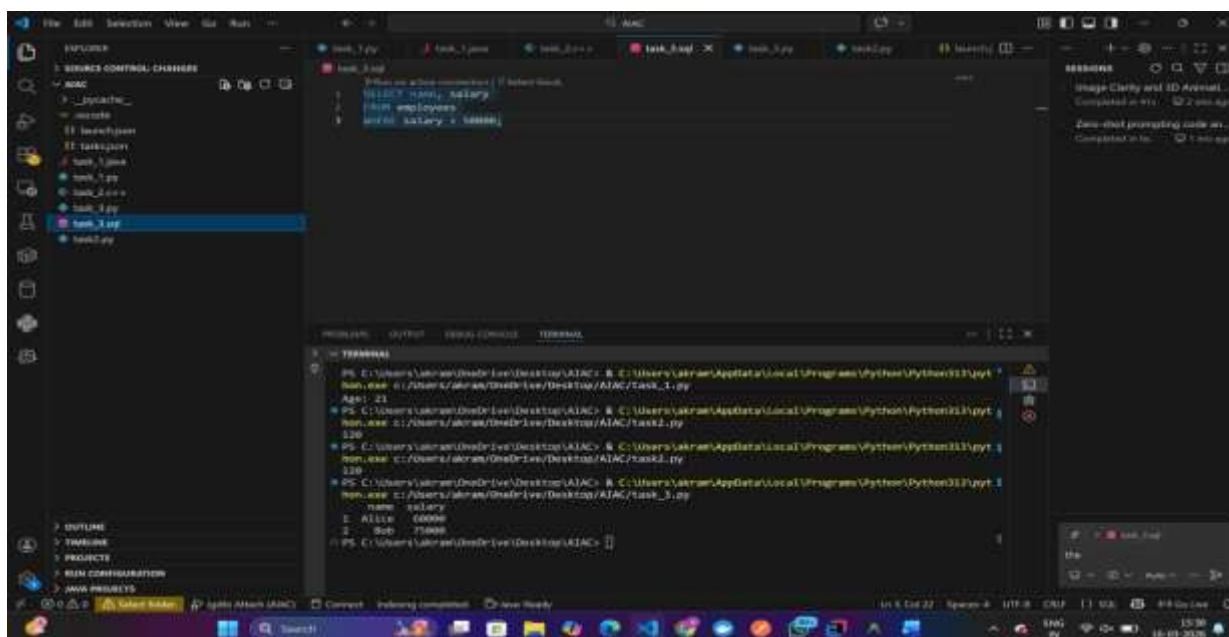
- Equivalent Pandas query that retrieves the correct subset of data.

CODE:



```
task_1.py
1 import pandas as pd
2
3 data = {
4     'name': ['Alice', 'Bob', 'Tom'],
5     'salary': (40000, 60000, 70000, 40000)
6 }
7
8 df = pd.DataFrame(data)
9
10 result = df[df['salary'] > 50000][['name', 'salary']]
11
12 print(result)
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python311\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_1.py
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python311\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python311\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_3.py
name salary
2 Bob 70000
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```



```
task_3.py
1 # Filter out those employees who have salary
2 from employees
3 where salary > 50000
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python311\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_1.py
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python311\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python311\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_3.py
name salary
1 Alice 60000
2 Bob 70000
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

for browser automation.

- Translating a Java method into C# for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least two different programming languages

CODE:

